

Tseegi Nyamdorj

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Education

Smith College, Northampton MA
B.A, Computer Science & Mathematical Statistics
GPA: 3.84/4.00, Dean's List 2021–2022

Sep 2021–May 2025

Awards / Honors

Middlebury College Project for Peace (\$10,000) - 2022
Smith College Universal Conference Fund (\$400) to attend CMSAC 2023
The Novice Champion in the 10th Asian Schools Debating Championship (ASDC) in Bangkok, Thailand - 2019
The Medal of Excellence issued by the Mongolian Government for outstanding study - 2019

Research Experience

Smith College – Department of Computer Science

May 2024–Present

Undergraduate Research Assistant

- Conducted research in document engineering with Professors Halie Rando and Michael Robson.
- Developed a LLM pipeline using Llama3 for integrating track changes made in DOCX files into reproducible, collaborative manuscript writing workflows in Manubot (an open-source software project built on Markdown).
- Optimized model selection and fine-tuned prompt engineering for increased speed and accuracy with Promptfoo.
- Utilized Python, LLMs, and NLP techniques to automate document conversion.
- Presented research at the Computational Systems for Integrative Genomics (CSIG) 2024.

Carnegie Mellon University – Department of Statistics & Data Science

May 2023– August 2023

Undergraduate Researcher

- Conducted statistical research in sports analytics with Professor Ron Yurko.
- Developed a machine learning model using R to predict soccer match outcomes by analyzing betting and player performance metrics.
- Evaluated model performance with Brier score and calibration plots to improve prediction accuracy.
- Applied advanced techniques in data preprocessing, feature selection, and model evaluation to sports analytics.
- Presented results at the Carnegie Mellon Sports Analytics Conference (CMSAC) 2023.

Personal Projects

CacheBite - CSC 256 Course Project

Feb 2024–Present

- Designed an app to connect students needing assistance (e.g., meal deliveries, mail pick-ups) with peers willing to help.
- Conducted data collection through surveys, observations; performed analysis using R.
- Developed user interfaces using XCode and Figma; delivered a functional demo on XCode.
- Addressed critical accessibility gaps, promoting inclusivity and connection on campus.

Minimax ChessBot on CUDA - CSC 220 Course Project

Sep 2024–Present

- Implemented a Minimax chessbot with alpha-beta pruning in C++ and CUDA.
- Utilized the SHL chess library to streamline board representation and move generation.
- Optimized computational efficiency by $5.7\times$ speed-up compared to Python-based implementations by parallelizing Minimax calculations.
- Delivered a functional chessbot capable of evaluating and making strategic moves efficiently.

Adults Pursuing Higher Education - SDS 291 Course Project

Sep 2024–Present

- Analyzed factors influencing adult education using logistic regression on IPUMS data
- Conducted data wrangling and visualization in R to explore relationships between age, parent-hood, and educational enrollment.
- Developed and compared multiple statistical models to identify significant predictors, achieving robust hypothesis testing through chi-squared analysis.

Other Experience**Smith College – Department of Mathematical Sciences**

Sep 2022–Present

Teaching Assistant/ Course Grader

- Evaluated assignments and projects in Calculus I-III, Discrete Mathematics, and Mathematical Statistics seminar
- Provided feedback on student work and answered course-related questions

AFS Mongolia

Jul 2020–Sep 2023

Systems Manager

- Increased website SEO ranking by 207.3% using Google Search Console, Analytics, Tag Manager, and HTML updates.
- Implemented GDPR compliance protocols as a certified Data Protection Officer.
- Led technical support for platforms: Salesforce, Trello, and Workplace.
- Managed a cohort of 10-15 summer interns on a weekly basis; conducted technical orientations.

Food Rescue Network – Smith College

Oct 2021 – Dec 2023

Vice-President

- Spearheaded the founding and growth of the FRN. Led club activities and facilitated teamwork among board members.
- Analyzed food collection, weather patterns, and food surplus in Python, to develop food waste prevention reports for dining halls at the end of each semester.
- Reduced and helped redirect 15k pounds of food waste on campus in the first two years.
- Developed leadership skills through strategic planning, conflict resolution, and mentoring of other board members.

Joseph Metcalf High School – Mindich Fellowship

Jan 2022–Feb 2022

Teaching Assistant

- Selected to a cohort of Mindich fellows that engage in school-based teaching practica during the one-month apprenticeship
- Prepared social/ethnic studies and mathematics teaching materials for 6th graders and helped mentor teacher during classroom and recess

Conference Presentations**Talks**

- [T-1] To Bet or Not to Bet: Analyzing the Credibility of Fixed-odds Betting on Match Outcomes.
– At the 5th Carnegie Mellon Sports Analytics Conference (**CMSAC**), Pittsburgh, PA, USA, November 10-11, 2023.
- [T-2] Conversion of Docx to Manubot-based Markdown file
– At the Computational Systems for Integrative Genomics (**CSIG**), New York, NY, USA, July 29-30, 2024.

Posters

- [P-1] Fungai Jani, Tsengelmaa Nyamdorj, Maria Tsakalakos. To Bet or Not to Bet: Analyzing the Credibility of Fixed-odds Betting on Match Outcomes. In *Carnegie Mellon Sports Analytics Conference*, 2023 at Carnegie Mellon University